Experiment 8: Ore prepration - Lab Report

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1) Aim

This experment examines how an ore is prepared and upgraded. The Specific aims are to grind a galena (PbS) ore, Determine its particle Size distribution by sieve analysis, plot the cumulative oversize and undersize Curves to find the average particle size, and compare it with the theoritical value. Furthermore, it aims to concentrate galena by froth flotation and Calculate the enrichment ratio(z)

2) Materials and Devices

- Galena (PbS)
- 1 Liter of water
- Collector(1mg) · Frothar (1drop) — mixed.
- Planetary type mill, test sieves, sieving machine, Denver flotation machine, precision balance, drying oven (105°C).

3) Experimental Procedure

3.1) Ore preparation and Sieves Analysis

- Exactly 1000g of galena Ore was weighed and ground in planetary mill for 10 munites at high energy.
- The test sieves were stacked from largest (4mm) to smalest (0.18mm) mesh number.
- Ground ore was fed onto the top sieve clamped, and the machine was run for 10 minutes.
- The mass retained on each sieve was weighed and recorded.

3.2) Flotation Experiment

- A pulp was prepared by mixing 200g of the ground ore with 1 Liter of water in the flotation cell. The pulp was stirred for 5min.
- A trace amount of collector (1mg) was added to make the galena hydrophobic and stirring continued for another 5 minutes.

- One drop of frother was added and pulp was stirred for 1-2 more minutes to generate froth.
- The PbS-rich froth rising to the surface was skimmed into a seperate container.
- The collected liquid froth was dried in an oven at 100-105°C and weighed to determine the concentrate mass.

4) Data Derived from the Experiment.

Table 1. Sieve Analysis of the Ground ore

Sieve Opening (mm)	Weight (g)	Weight (%)	Cumulative oversize (%)	Cumulative Undersize (%)
4.00	129	12.9	12.9	87.1
2.83	251	25.1	38.0	62.0
2.36	44	4.4	42.4	57.6
1.20	245	24.5	66.9	33.1
0.50	95	9.5	76.4	23.6
0.25	36.5	3.65	80.05	19.95
0.18	25	2.5	82.55	17.45
0.09 (undersize)	174.5	17.45	100.0	0.0
total	1000	100.0		

- Theoretical avarage particle size $\frac{\Sigma(x.m)}{100}$: 1,70 mm
- Avarage particle size from curve instrection: ≈ 2.00 mm

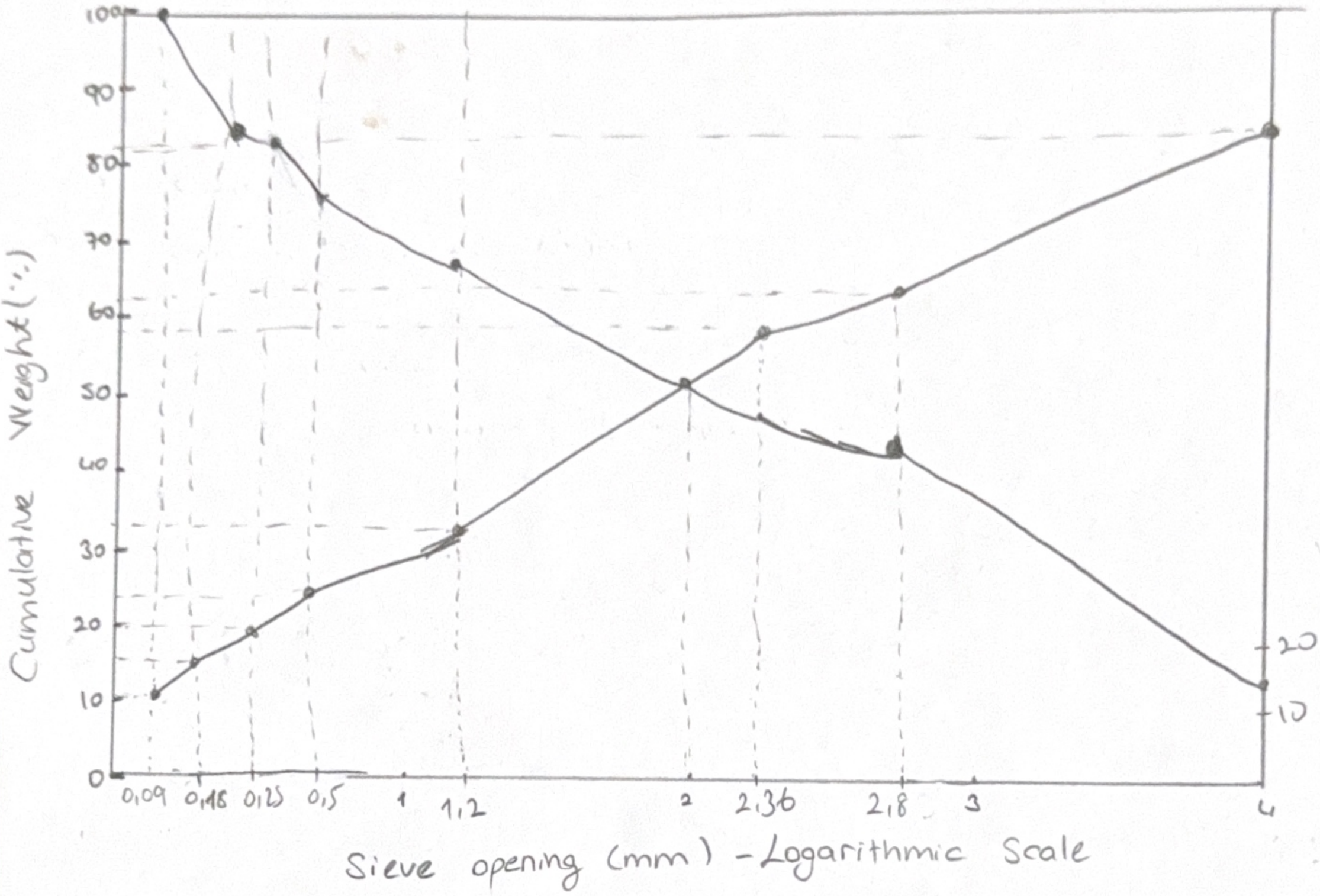


Table 2. Flotation Mass Balance

Stream	mass	• Enrichment Ratio (Z) =
Feed (ground ore)	200	
Concentration	13	$= \frac{\text{feed mass}}{\text{concentration mass}} = \frac{200}{13} \approx 15.38$
Taillings	187	

5) Discussion and Evaluation of the Results.

The experment confirmed that grinding liberates galena and flotation effectively concentrates it. In sieve analysis, the graphical particle size (2.00mm) differed slightly from the theoretical 1.70 mm. This variance stems from interpolation limits and a high fines fraction (%17.45) pulling the calculated allorage down.

During flotation, a trace collector (1mg) yielded 13 g of concentrate from 200 g of feed, achieving a high enrichment ratio of 15.4. Assuming a %6 initial feed grade and %80 recovery, the final grade up-graded to ~73,8 %. (12,3-fold enrichment). Minor mass discrepancies suggest slight material losses as dust, during frath skimming or from incomplete drying.

6) References

- Erguvanlı, F., "cevher hazırlama prensipleri", İTÜ Publications, 1959.
- Gizmecioğlu, Z., Üretim metalurjisi Prensipleri, YTÜ, 2008
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- Kaytoz, Y., "Cevher Hazırlama", İTÜ Facility of Mining, 1990.